Health Risk Prediction Under Distribution Shift in Heterogeneous Patient Populations

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Abstract

Machine learning models for cardiac risk prediction routinely fail to generalize across heterogeneous patient populations, yet this degradation is rarely quantified under controlled experimental conditions. This paper investigates out-of-distribution (OOD) generalization under distribution shift by training seven classifiers on the UCI Heart Disease Multi-Center dataset and evaluating them on the Framingham Heart Study — two datasets with fundamentally different feature spaces, demographic compositions, and class distributions (55.3% vs. 15.2% positive rate). Exploratory analysis confirmed severe multi-dimensional shift: statistically significant covariate shift across all shared features and label shift of 3.6× confirmed via chi-squared test . In-domain models achieved ROC-AUC up to 0.904; OOD performance degraded to 0.657–0.685, with F1 collapsing from ∼0.85 to ∼0.33. Five adaptation strategies were evaluated: hyperparameter tuning, class weighting, SMOTE, threshold tuning, and pooled training. No strategy resolved the OOD gap; class weighting achieved the best F1 (0.361), threshold tuning maximized recall (0.744), and pooled training optimized precision (0.413) at the cost of recall. Results demonstrate that source-domain validation metrics are unreliable indicators of real-world performance, and that structural feature gaps and label shift require target-domain-aware mitigation strategies for responsible clinical deployment.

1 Introduction and Motivation

Cardiovascular diseases (CVDs) are a leading cause of death worldwide, accounting for approximately 17.9 million deaths annually (32% of global deaths). In 2021, coronary heart disease alone caused around 9 million deaths globally. The burden is expected to increase significantly, with 61% of U.S. adults projected to be hypertensive by 2050, and global CVD deaths rising from 20.5 million in 2025 to 35.6 million by 2050. These figures highlight the urgent need for improved early detection and prevention strategies (Kumar et al., 2025).

Traditional diagnostic approaches for heart disease rely on laboratory testing, medical imaging, and clinical expertise. While these methods are widely used in practice, they face limitations when dealing with large-scale and complex medical data. With the rapid growth of electronic health records (EHRs), healthcare systems now generate vast amounts of patient data that are difficult to analyze efficiently using conventional techniques. In addition, human-based diagnosis may introduce subjectivity due to variations in clinical judgment (Cai, 2024).

Recent advances in machine learning (ML) have provided powerful tools for addressing these challenges. ML algorithms can process large volumes of structured and unstructured data, identify hidden patterns, and generate predictive insights based on patient information. As a result, ML has become increasingly important in healthcare applications such as early disease detection, risk assessment, and clinical decision support (Hossain, 2024).

Several studies have demonstrated the effectiveness of machine learning models in predicting heart disease. Algorithms such as Logistic Regression, Random Forest, and Support Vector Machines (SVM) have achieved strong performance in classification tasks and improved diagnostic accuracy (Alsabhan and Alfadhly, 2025; Bouqentar, 2024). These models enable data-driven decision-making and have the potential to enhance the efficiency and reliability of healthcare systems (Teja, 2025).

Despite these advancements, challenges remain in developing robust and generalizable models. In particular, variations in data quality, missing values, and differences between datasets can significantly affect model performance. Ensuring that models generalize well across different patient populations is a key concern in medical machine learning appli-

cations (Cai, 2024).

2 Problem Definition

Heart disease prediction can be formulated as a supervised binary classification problem, where the objective is to classify individuals into two categories: patients with heart disease and those without. This classification is based on a set of input features derived from patient demographics and medical records.

Common features include age, cholesterol levels, blood pressure, and heart rate, all of which are known to be important indicators of cardiovascular health. For example, high cholesterol levels can lead to plaque buildup in arteries, increasing the risk of heart disease, while elevated blood pressure affects blood flow and cardiovascular function (Kumar et al., 2025).

In this context, machine learning models are trained on historical patient data to learn the relationship between these features and the presence of heart disease. Once trained, these models can be used to predict the likelihood of disease in new patients. Various algorithms, including decision trees, k-nearest neighbors, support vector machines, and neural networks, have been applied to this task with varying levels of complexity and performance (Bouqentar, 2024).

Model performance is typically evaluated using metrics such as accuracy, precision, recall, and F1-score to ensure reliability in medical decision-making (Hossain, 2024). However, challenges such as missing data, class imbalance, and dataset variability must be addressed to develop effective predictive models (Cai, 2024).

A central challenge in this work is ensuring that models trained on one patient population generalize to a demographically and clinically distinct target population — a problem of out-of-distribution (OOD) generalization under distribution shift.

3 Related Work

3.1 Machine Learning Approaches for Heart Disease Prediction

Machine learning methods have become indispensable in the early detection and diagnosis of cardiovascular diseases. By leveraging large-scale and heterogeneous patient data, ML models assist clinicians in predicting disease risk with higher precision and objectivity than traditional diagnostic techniques. Recent studies underscore the superiority of both classical and advanced ML approaches in improving diagnostic performance and enabling population-level generalization (Kumar et al., 2025; Alsabhan and Alfadhly, 2025; Himanshu and Meenu, 2025).

3.1.1 Traditional Classification Models

Logistic Regression has been one of the earliest models applied to heart disease prediction due to its interpretability and efficiency on small to moderate datasets. It maps input features such as age, cholesterol, blood pressure, and fasting blood sugar to a probabilistic estimate of disease risk. Recent comparative studies report accuracies ranging from 85%–92%, depending on dataset size and preprocessing quality (Bouqentar, 2024; Teja, 2025). Regularized variants such as L1 (LASSO) and L2 (Ridge) mitigate overfitting and improve generalization on sparse healthcare data (Hossain, 2024).

Decision Trees and Random Forests provide intuitive hierarchical decision rules through recursive feature partitioning, though standalone trees are prone to overfitting on noisy health datasets. Ensemble-based Random Forests combine multiple trees via bagging to reduce variance and enhance robustness. A 2025 comparative study reports RF achieving 98.1% accuracy and 97.5% sensitivity on the UCI Cleveland dataset (Teja, 2025). Feature importance measures from RF also provide clinical interpretability, allowing practitioners to understand variable contributions — such as max heart rate and ST depression — to disease outcomes.

Naïve Bayes and K-Nearest Neighbors perform competitively in low-dimensional settings with limited preprocessing needs. Naïve Bayes handles categorical variables effectively under small-sample regimes, while KNN achieved over 88% accuracy on the UCI dataset using optimized scaling and distance metrics (Teja, 2025). However, scalability issues limit KNN's utility on high-dimensional EHR data.

3.1.2 Advanced and Ensemble Methods

Support Vector Machines are widely adopted in cardiovascular prediction tasks due to their ability to separate classes in non-linear domains using kernel transformations. Studies demonstrate that SVM achieves superior recall and F1-scores compared to classical models when suitable kernel and scaling choices are applied (Kumar et al., 2025; Bouqentar, 2024). Gaussian and RBF kernels achieve precision rates exceeding 90% in benchmark studies

(Silva et al., 2025).

Gradient Boosting and XGBoost represent state-of-the-art ensemble learners that iteratively minimize classification error by combining weak decision trees. XGBoost has demonstrated leading performance across multiple benchmark datasets, with accuracies up to 97.6% and AUC-ROC values of 0.98 on the Cleveland dataset (Singh et al., 2022). These models handle feature interactions and non-linearities effectively and prove robust under moderate class imbalance through built-in regularization and tree-weighting mechanisms.

Deep Neural Networks and Hybrid Models extend traditional ML through architectures including MLPs, CNNs, and LSTMs. Hybrid CNN-LSTM frameworks capture spatial and temporal dependencies from clinical time-series data such as ECG signals, achieving 99.38% accuracy and 98% AUC on extended heart disease datasets (Singh et al., 2022). Hybrid ensemble approaches combining RF, XGBoost, and neural network outputs through weighted voting schemes achieve a 3–15% performance boost over single classifiers (Singh et al., 2022).

3.1.3 Performance Trends in Literature

Across comparative studies, ensemble and deep learning models generally outperform standalone classifiers. XGBoost, CNN-LSTM, and Random Forest consistently deliver the best trade-offs between accuracy ($\geq$95%), precision ($\geq$92%), and AUC ($\geq$0.97) (Kumar et al., 2025; Alsabhan and Alfadhly, 2025; Himanshu and Meenu, 2025; Bouqentar, 2024; Teja, 2025; Musanga et al., 2026). However, interpretability remains limited in deep architectures, prompting integration of Explainable AI techniques such as SHAP and LIME for clinical transparency (Kumar et al., 2025; Wiens and Shenoy, 2018).

3.2 Heterogeneous Healthcare Data and Multi-Center Datasets

This project utilizes two primary datasets: the UCI Heart Disease Multi-Center dataset, aggregating clinical records from five hospitals across Cleveland, Hungary, Switzerland, VA Long Beach, and Statlog (approximately 1,190 patients), and the Framingham Heart Study, a longitudinal cohort tracking 4,434 patients over time. Both datasets include shared cardiovascular risk factors — age, blood pressure, cholesterol, and heart rate — while differing substantially in feature space, study design, and population composition.

Combining datasets from multiple sources is beneficial because it brings together information from different patient groups and clinical settings, making models more adaptable to real-world heterogeneity. Prior work has applied Logistic Regression, Decision Trees, and Neural Networks to such combined datasets, typically following pipelines of data cleaning, missing value handling, feature scaling, and evaluation via accuracy, precision, recall, and F1-score (Chen et al., 2021).

Characteristics of Multi-Center Data. Multi-center datasets exhibit systematic differences across hospitals due to variations in data collection protocols, equipment, and recording practices. These differences affect the representation of certain attributes and directly impact model learning (Wiens and Shenoy, 2018).

Population Diversity. The datasets contain patients spanning diverse age groups, demographics, and clinical conditions. Diversity in training data is essential for reducing demographic bias and improving model generalization to future populations (Subbaswamy and Saria, 2020).

Challenges in Combining Datasets. Merging datasets introduces challenges including feature inconsistency — where the same variables are represented differently across sources — and variable data quality, such as missing or noisy entries. Preprocessing must address these issues prior to model training to prevent accuracy degradation (Ristevski and Chen, 2018).

3.3 Distribution Shift in Machine Learning

3.3.1 Definition

Distribution shift occurs when the statistical characteristics of training data differ from those encountered during testing or real-world deployment. A model trained on source distribution $P(X, Y)$ will experience performance degradation when the target distribution $Q(X, Y)$ diverges from P, violating the i.i.d. assumption of supervised learning. In healthcare, this is especially critical because models can degrade silently — internal cross-validation uses holdout data drawn from the same source distribution and therefore fails to expose mismatches that emerge in real-world deployment (Blaas et al., 2024).

3.3.2 Types of Distribution Shift

Covariate shift occurs when $P(X) \neq Q(X)$ while $P(Y|X) = Q(Y|X)$ — the input feature

distributions change but the feature-outcome relationship is preserved. This arises frequently in clinical settings from demographic differences between patient cohorts across hospitals (Silva et al., 2025).

Label shift occurs when $P(Y) \neq Q(Y)$ while class-conditional feature distributions remain unchanged. Disease prevalence varies across populations and over time, causing models trained on one prevalence rate to be miscalibrated when deployed in a different setting. This is directly relevant to our experimental setting, where the positive rate drops from 55.3% (UCI) to 15.2% (Framingham) (Silva et al., 2025; Gardner et al., 2023).

Concept drift describes changes in $P(Y|X)$ — the relationship between features and outcomes evolves over time due to shifts in treatment protocols, clinical guidelines, or diagnostic criteria. Unlike covariate shift, concept drift fundamentally invalidates learned decision boundaries and cannot be corrected through sample reweighting alone (Gardner et al., 2023; Guo et al., 2021).

Domain shift is the overarching challenge that subsumes covariate shift, label shift, and concept drift. It encompasses all differences in EHR implementation, data collection standards, clinical coding practices, and institutional processes between source and target environments (Yoon et al., 2024). Large performance discrepancies consistently emerge when models trained on one population are applied to a demographically or institutionally different target (Musa et al., 2025; Singh et al., 2022).

Subdomain shift refers to performance degradation concentrated in specific demographic or clinical subgroups underrepresented in training data, rather than manifesting uniformly across the target population (Singh et al., 2022). Mortality risk models have exhibited performance disparities across racial groups even while maintaining strong aggregate metrics, indicating that population-level performance statistics alone are insufficient to characterize model robustness in heterogeneous populations (Singh et al., 2022).

3.3.3 Impact on Model Performance

Distribution shift degrades both generalization and deployment reliability. ML-based clinical risk models are highly sensitive to distributional changes, and AI models can suffer severe performance decay over years of deployment due to temporal dataset shifts (Silva et al., 2024; Guo et al., 2023). A prognostic study across 143,049 patients at seven Toronto hospitals identified simultaneous multi-dimensional shifts across demographics, hospital types, admission sources, and laboratory assays (Subasri et al., 2025) — illustrating how distribution shift can be both multi-dimensional and silent without dedicated monitoring infrastructure. Shift also produces biased predictions that disproportionately affect underrepresented subgroups; a 2024 review found that 73% of clinical text datasets originated from the Americas and Europe, regions representing only 22% of the global population (Yang et al., 2024).

3.4 Methods for Handling Distribution Shift

Domain Adaptation adapts a source-trained model to a related target domain without requiring labeled target data. Feature-level adversarial adaptation learns domain-invariant representations, enabling site-independent performance without retraining (Musanga et al., 2026). Drift-triggered continual learning strategies have also been shown to reduce harmful data shifts over deployment periods (Subasri et al., 2025).

Transfer Learning reuses knowledge from a source task as initialization for a related target task. Self-supervised EHR foundation models pretrained on large diverse corpora demonstrate improved robustness and reduced performance decay relative to standard baselines (Guo et al., 2023). A multi-center study further showed that adapting EHR foundation models across hospitals yields improved prediction at reduced annotation cost (Guo et al., 2024).

Data-Level Techniques adjust training data composition to better reflect target distributions. Approaches range from SMOTE and ADASYN oversampling to GAN-based synthetic data generation (Marchesi et al., 2025). Class weighting assigns higher penalties to misclassified minority samples, particularly effective in Logistic Regression and SVM (Marchesi et al., 2025).

Model Robustness Strategies include training on diverse datasets and federated learning, which enables multi-institutional model development without direct data sharing (Pan et al., 2023; Teo et al., 2024). Explicitly incorporating causally stable covariates into prediction models has also been shown to recover robustness under covariate shift (Blaas et al., 2024).

4 Data Description and Preprocessing

4.1 Datasets

UCI Heart Disease Multi-Center Dataset aggregates clinical records from five institutions (Cleveland, Hungary, Switzerland, VA Long Beach, Statlog), comprising 918 patient records after deduplication. Features include 11 clinical variables: age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, ST depression (oldpeak), and ST slope. The target variable indicates presence of heart disease (55.3% positive).

Framingham Heart Study is a longitudinal community cohort comprising 4,240 records. Features include 15 epidemiological variables: age, sex, smoking status, cigarettes per day, blood pressure medication, prevalent stroke, prevalent hypertension, diabetes, total cholesterol, systolic and diastolic blood pressure, BMI, heart rate, glucose, and education. The target variable indicates 10-year coronary heart disease risk (15.2% positive).

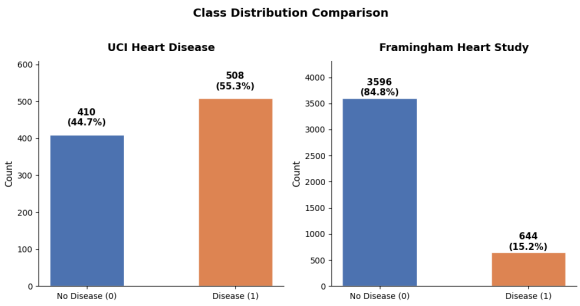

Figure 1: Class distribution comparison between UCI (55.3% positive) and Framingham (15.2% positive), illustrating the 3.6× label shift motivating OOD evaluation.

4.2 Feature Alignment

The two datasets share no identical feature set, requiring a two-layer evaluation strategy. **Layer A** uses each dataset's full feature space for in-domain evaluation. **Layer B** constructs a shared feature set of four aligned variables — `age`, `sex`, `cholesterol`, and `blood_pressure` — for cross-domain OOD evaluation. Sex encoding and column names were standardized across datasets prior to evaluation.

4.3 Preprocessing

UCI cholesterol contained 172 zero-encoded missing values (physiologically impossible), imputed with the column median (236.5 mg/dL). One zero blood pressure value was similarly corrected. Categorical features (chest pain type, resting ECG, ST slope) were one-hot encoded. All continuous features were standardized using z-score normalization within pipeline steps to prevent data leakage. No rows were dropped; dataset shapes were preserved at 918 and 4,240 observations respectively.

4.4 Domain Shift Evidence

Kolmogorov-Smirnov tests confirmed statistically significant distributional differences across all shared features (age: KS=0.212, p=4.1e-30; cholesterol: KS=0.125, p=9.8e-11; blood pressure: KS=0.127, p=3.9e-11). A chi-squared test confirmed severe label shift (χ^2=698.94, p=5.1e-154), with the positive rate differing by a factor of 3.6× across datasets. These results establish the distributional mismatch that motivates all subsequent experimental phases.

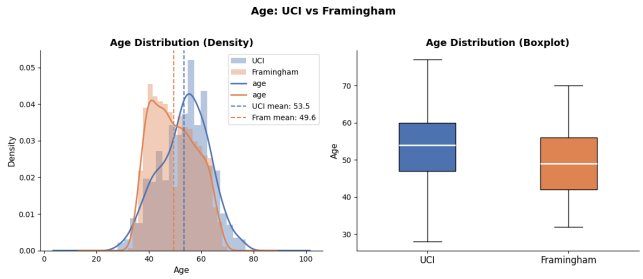

Figure 2: Age distributions across UCI and Framingham. UCI patients are systematically older (mean 53.5 vs. 49.6), confirmed by KS test ($p = 4.1 \times 10^{-30}$).

5 Methodology

5.1 Experimental Design

We adopt a source-to-target transfer evaluation protocol: all models are trained exclusively on UCI data and evaluated on Framingham as an out-of-distribution (OOD) test set, simulating real-world deployment across heterogeneous patient populations. A two-layer feature strategy governs evaluation. Layer A uses each dataset's full feature space for in-domain baseline establishment. Layer B restricts training and testing to four shared features (`age`, `sex`, `cholesterol`, `blood_pressure`) for cross-domain OOD evaluation. All cross-domain experiments operate exclusively on Layer B to ensure feature space consistency between train and test.

The UCI dataset is split 80/20 for in-domain training and validation. Framingham is held out

entirely as the OOD test set across all phases except Phase 7 (pooled training), where a portion of Framingham is incorporated into training. Evaluation metrics — Precision, Recall, F1-score, and ROC-AUC — are reported for all phases. Accuracy is excluded due to severe class imbalance between datasets (55.3% vs. 15.2% positive rate).

5.2 Models

Seven classifiers are evaluated: Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naïve Bayes (NB), AdaBoost, and Gradient Boosting (GB). This selection spans linear, kernel-based, instance-based, probabilistic, and ensemble paradigms, enabling a comprehensive assessment of how model family affects OOD robustness. All distance-sensitive models (LR, SVM, KNN) are wrapped in standardization pipelines to prevent feature-scale leakage.

5.3 Mitigation Strategies

Five mitigation strategies are applied sequentially to assess their effectiveness under distribution shift:

Hyperparameter tuning applies 5-fold cross-validated GridSearchCV on the UCI training set, optimizing ROC-AUC for LR, RF, AdaBoost, and GB. This establishes whether source-domain optimization improves OOD transferability.

Class weighting applies `class_weight='balanced'` to LR and SVM, inversely weighting classes by frequency to partially correct for the 3.6× label shift between datasets.

SMOTE generates synthetic minority-class samples to balance the UCI training distribution, applied to LR, RF, and GB prior to OOD evaluation.

Threshold tuning sweeps decision thresholds from 0.10 to 0.90 on the UCI validation set and selects the threshold maximizing F1. The optimal threshold is then applied directly to Framingham probability outputs, requiring no target-domain labels.

Pooled training combines the UCI training set with the full Framingham dataset (excluding a 30% holdout) to directly expose models to the target distribution during training, evaluating LR, RF, and GB.

6 Experimental Results and Discussion

6.1 In-Domain Baseline

Seven classifiers trained on the full UCI feature set (Layer A) achieved strong in-domain performance. AdaBoost led with ROC-AUC of 0.904 and F1-score of 0.859, followed by Gradient Boosting (AUC 0.891, F1 0.844) and SVM (AUC 0.887, F1 0.847). These results establish the performance ceiling under source-domain conditions and serve as the reference against which OOD degradation is measured.

6.2 OOD Performance Degradation

When evaluated on Framingham using only the four shared features (Layer B), all models experienced severe degradation. ROC-AUC dropped to 0.657–0.685 and F1-score collapsed to 0.322–0.348 — a reduction of approximately 0.50 F1 points relative to in-domain performance. Precision fell to 0.24–0.28 across all models, reflecting systematic over-prediction of the positive class caused by the 3.6× label shift between datasets (55.3% UCI vs. 15.2% Framingham). AdaBoost achieved the strongest OOD baseline (F1=0.348, AUC=0.685); Random Forest ranked last (F1=0.322, AUC=0.657).

6.3 Adaptation Strategy Comparison

Table 1 reports the best OOD result achieved by each adaptation strategy. Class weighting produced the largest F1 improvement over baseline (+0.013), offering a practical correction for label shift without requiring target-domain data. Threshold tuning achieved the highest recall (0.744) — critical in clinical settings where false negatives carry greater cost than false positives — at the expense of precision. Hyperparameter tuning and SMOTE produced negligible gains (ΔF1 $\leq$ 0.001), confirming that source-domain optimization does not address distributional mismatch. Pooled training substantially degraded F1 (ΔF1 = -0.106), as naive combination of heterogeneous source and target data introduced conflicting patterns rather than improving generalization; however, it achieved the highest precision (0.413), representing a different operational tradeoff.

Strategy	Best Model	Precision	F1	ΔF1	Recall	AUC
Baseline OOD	AdaBoost	0.277	0.348	—	0.467	0.685
Class Weighting	SVM (bal.)	0.272	0.361	+0.013	0.488	0.683
SMOTE	GB + SMOTE	0.261	0.335	-0.013	0.450	0.688
Hyper. Tuning	GB (tuned)	0.248	0.349	+0.001	0.561	0.677
Threshold Tuning	RF	0.225	0.345	-0.003	0.744	0.688
Pooled Training	LR	0.413	0.242	-0.106	0.171	0.693

Table 1: Best out-of-domain performance per adaptation strategy. ΔF1 relative to OOD baseline (AdaBoost, F1=0.348). Pooled training achieves the highest precision (LR: 0.413) at the cost of recall.

6.4 Discussion

A consistent finding across all phases is that ROC-AUC remains stable (0.657–0.693) regardless of strategy, indicating that domain shift degrades decision-boundary calibration more than ranking ability. No single strategy dominates across all metrics. The optimal choice depends on the clinical deployment objective: threshold tuning is preferred when minimizing missed diagnoses is the priority; pooled training when precision is critical; class weighting when a balanced tradeoff is required under source-only conditions.

The marginal gains achieved by all strategies underscore a fundamental limitation: when label shift is severe and feature gaps are structural, source-side interventions provide limited relief. Meaningful OOD robustness likely requires access to representative target-domain data at training time or more sophisticated domain adaptation techniques beyond the scope of this work (Subbaswamy and Saria, 2020; Silva et al., 2025).

7 Conclusion

This paper investigated the generalization behavior of binary cardiac risk classifiers trained on the UCI Heart Disease dataset and evaluated on the Framingham Heart Study under controlled out-of-distribution conditions. The experimental pipeline systematically quantified the cost of distribution shift and evaluated five practical mitigation strategies within a standard supervised learning framework.

Domain shift is severe and multi-dimensional. Statistically confirmed covariate shift across all shared features, a $3.6\times$ label prior difference, a structural feature gap between datasets, and divergent correlation structures collectively created a challenging transfer scenario that no single mitigation strategy fully resolved.

Strong in-domain performance does not imply OOD robustness. Models achieving ROC-AUC of 0.90+ on UCI validation degraded to 0.66–0.69 on Framingham — a gap that would be invisible under standard source-only cross-validation, underscoring the risk of evaluating clinical models exclusively on source-domain holdout sets.

Source-side interventions provide limited relief. Hyperparameter tuning and SMOTE produced negligible OOD gains ($\Delta F1 \leq 0.001$), confirming that optimizing the source distribution more precisely does not improve transferability. Class weighting achieved the most favorable overall tradeoff (F1=0.361), while threshold tuning maximized recall (0.744) and pooled training maximized precision (0.413) — each representing a different operational point rather than a global improvement.

Practical implication. When deploying cardiac risk models across heterogeneous patient populations, source-domain validation metrics are unreliable indicators of real-world performance. Awareness of label distribution differences and evaluation on representative target-domain data are essential prerequisites for responsible clinical deployment (Subbaswamy and Saria, 2020; Silva et al., 2025).

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